

Department of Mechanical and Production Engineering

Aarhus University

Exam

Course: Statics & Strength of Materials — 290221U006

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Problem 1

Consider the situation shown in Fig. 1a, where a square plate (in yellow) is connected to a wall via a ball and socket joint at point A and by a smooth support at point B . The ball and socket joint prevents translation in all directions, whereas the smooth support prevents translation along the negative y axis. To maintain equilibrium, the plate is also connected to two cables at points C and D . The thickness of the plate can be assumed uniform and negligible in relation to the other geometrical dimensions. Therefore, points A , B , C , and D can be considered to all lie on the x - z plane.

A composite body is placed on the plate. The body consists of a square block and a cylinder placed on top of each other. The block, shown in dark blue in Fig. 1b, is made of steel and has section measuring $1\text{ m} \times 1\text{ m}$ and a height of $0,5\text{ m}$. The cylinder, shown in light blue, is made of glass and has a diameter of $0,4\text{ m}$ and a height of $0,8\text{ m}$. The centroids of the block and the cylinder lie on an axis parallel to the y axis, which intersects the center of the plate.

- Q1.1 Determine the coordinates of the center of gravity of the composite body, assuming densities of 8000 kg/m^3 for steel and 2000 kg/m^3 for glass.
- Q1.2 Draw a free-body diagram of the plate indicating all the forces acting on it.
- Q1.3 Compute the unit vectors along which the tensions T_1 and T_2 from the cables connected at points C and D act. Express the unit vectors in Cartesian form.
- Q1.4 Use the equilibrium equations to find the magnitude of T_1 , T_2 and all the other reaction forces acting on the plate.

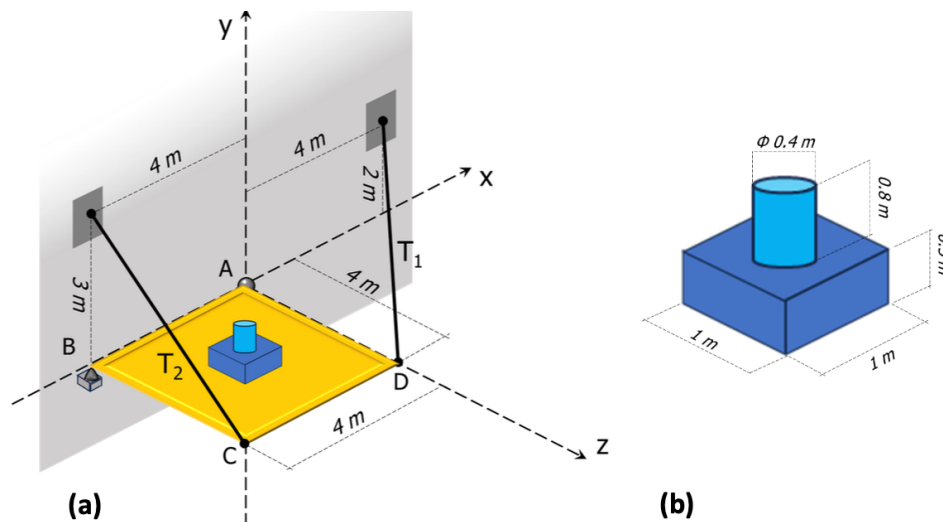


Fig. 1. (a) Schematic of the problem considered in Part 1. (b) Dimensions of the composite body placed on the plate.

Derivation

- Q1.1 In the xz plane, the center of gravity of the two parts of the composite body and the plate all lie at $(\bar{x}; \bar{z}) = (-2; 2)\text{ m}$. In the y direction the centroid of the composite body can be found using:

$$\bar{y} = \frac{\sum \bar{y}_i \cdot m_i}{m}.$$

Where $\bar{y}_i$ is the centroid location in the y direction of one of the parts of the composite body and m_i is the

mass of one part of the composite body whilst m is the mass of the entire composite body. Hence,

$$\begin{aligned}
 V_{\text{cyl}} &= \pi \cdot \frac{(0,4 \text{ m})^2}{4} \cdot 0,8 \text{ m} = 0,1 \text{ m}^3 \\
 V_{\text{block}} &= 1 \text{ m} \cdot 1 \text{ m} \cdot 0,5 \text{ m} = 0,5 \text{ m}^3 \\
 m_{\text{cyl}} &= 0,1 \text{ m}^3 \cdot 2000 \text{ kg/m}^3 = 200 \text{ kg} \\
 m_{\text{block}} &= 0,5 \text{ m}^3 \cdot 8000 \text{ kg/m}^3 = 4000 \text{ kg} \\
 \bar{y} &= \frac{0,25 \text{ m} \cdot 4000 \text{ kg} + 0,9 \text{ m} \cdot 200 \text{ kg}}{4200 \text{ kg}} = 0,28 \text{ m}.
 \end{aligned}$$

This means the coordinates of the center of gravity of the composite body are at:

$$(\bar{x}; \bar{y}; \bar{z}) = (-2; 0,28; 2) \text{ m}.$$

Q1.2 A free-body diagram is shown on [Figure 0.1](#). Here, we used the fact that A inhibits motion in all three axes whilst B only inhibits motion in the y axis. Furthermore:

$$W = mg = 4200 \text{ kg} \cdot 9,82 \text{ m/s}^2 = 41,244 \text{ kN}.$$

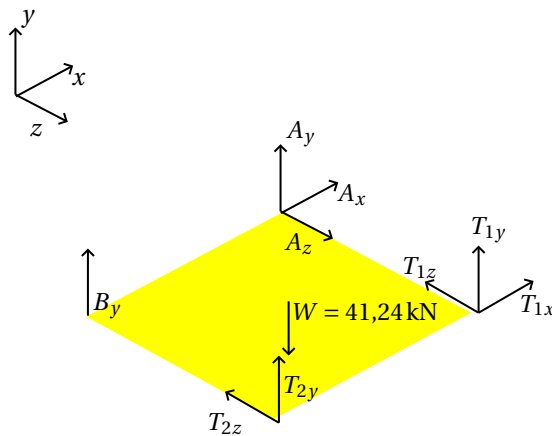


Figure 0.1: Free body diagram indicating the applied weight W and all reaction components and tension components.

Q1.3 We start by defining a vector from C to the point on the wall at which T_2 is fastened. I.e.

$$\mathbf{C}_{\text{wall}} = 0 \text{ m} \mathbf{i} + 3 \text{ m} \mathbf{j} - 4 \text{ m} \mathbf{k}.$$

To transform this into a unit vector we must divide it by its length:

$$\begin{aligned}
 \mathbf{u}_{T_2} &= \frac{\mathbf{C}_{\text{wall}}}{|\mathbf{C}_{\text{wall}}|} \\
 &= \frac{3 \text{ m} \mathbf{j} - 4 \text{ m} \mathbf{k}}{\sqrt{(3 \text{ m})^2 + (4 \text{ m})^2}} \\
 &= 0,6 \mathbf{j} - 0,8 \mathbf{k}.
 \end{aligned}$$

We can follow the same procedure for T_1 to get

$$\mathbf{D}_{\text{wall}} = 4 \text{ m} \mathbf{i} + 2 \text{ m} \mathbf{j} - 4 \text{ m} \mathbf{k}.$$

Which we divide by its length:

$$\begin{aligned}\mathbf{u}_{T_1} &= \frac{\mathbf{D}_{\text{wall}}}{|\mathbf{D}_{\text{wall}}|} \\ &= \frac{4\mathbf{m}\mathbf{i} + 2\mathbf{m}\mathbf{j} - 4\mathbf{m}\mathbf{k}}{\sqrt{(4\mathbf{m})^2 + (2\mathbf{m})^2 + (4\mathbf{m})^2}} \\ &= 0,666\mathbf{i} + 0,333\mathbf{j} - 0,666\mathbf{k}.\end{aligned}$$

Q1.4 A moment equilibrium about point A in the yz plane gives us

$$\begin{aligned}\sum (M_A)_{yz} &= 0 \\ T_{2y} \cdot 4\text{ m} - W \cdot 2\text{ m} + T_{1y} \cdot 4\text{ m} &= 0 \\ T_{1y} + T_{2y} &= \frac{41,24\text{ kN} \cdot 2\text{ m}}{4\text{ m}} \\ T_{1y} + T_{2y} &= 20,62\text{ kN} \\ T_1 \cdot 0,333 + T_2 \cdot 0,6 &= 20,62\text{ kN}.\end{aligned}$$

And a moment equilibrium about A in the xz plane gives

$$\begin{aligned}\sum (M_A)_{xz} &= 0 \\ T_{1x} \cdot 4\text{ m} - T_{2z} \cdot 4\text{ m} &= 0 \\ T_{1x} - T_{2z} &= 0 \\ T_1 \cdot 0,666 - T_2 \cdot 0,8 &= 0 \\ T_1 &= \frac{T_2 \cdot (0,8)}{0,666} = 1,2 T_2.\end{aligned}$$

Inserting this into our result from the moment equilibrium above we get:

$$\begin{aligned}1,2 T_2 \cdot 0,333 + T_2 \cdot 0,6 &= 20,62\text{ kN} \\ T_2 &= \frac{20,62\text{ kN}}{(0,6 + 1,2 \cdot 0,333)} \\ T_2 &= 20,62\text{ kN}.\end{aligned}$$

So

$$T_1 = 1,2 T_2 = 1,2 \cdot 20,62\text{ kN} = 24,74\text{ kN}.$$

So our tensions are:

$$T_1 = 24,74\text{ kN}$$

$$T_2 = 20,62\text{ kN}.$$

A force equilibrium in the x direction allows us to find A_x as

$$\begin{aligned}\sum F_x &= 0 \\ A_x + T_{1x} &= 0 \\ A_x &= -T_{1x} \\ A_x &= -24,74\text{ kN} \cdot 0,666 = -16,49\text{ kN}.\end{aligned}$$

A force equilibrium in the z direction gives us:

$$\begin{aligned}\sum F_z &= 0 \\ A_z - T_{1z} - T_{2z} &= 0 \\ A_z &= T_{1z} + T_{2z} \\ A_z &= 24,74\text{ kN} \cdot 0,666 + 20,62\text{ kN} \cdot 0,8\end{aligned}$$

$$A_z = 32,99 \text{ kN}.$$

And a force equilibrium in the y direction gives us:

$$\begin{aligned}\sum F_y &= 0 \\ A_y + B_y + T_{1y} + T_{2y} - W &= 0 \\ A_y + B_y &= W - T_{1y} - T_{2y} \\ A_y + B_y &= 41,24 \text{ kN} - 24,74 \text{ kN} \cdot 0,333 - 20,62 \text{ kN} \cdot 0,6 \\ A_y + B_y &= 20,62 \text{ kN}.\end{aligned}$$

A moment equilibrium about point A in the xy plane gives us

$$\begin{aligned}\sum (M_A)_{xy} &= 0 \\ -B_y \cdot 4 \text{ m} - T_{2y} \cdot 4 \text{ m} + W \cdot 2 \text{ m} &= 0 \\ B_y &= \frac{41,24 \text{ kN} \cdot 2 \text{ m} - 20,62 \text{ kN} \cdot 0,6 \cdot 2 \text{ m}}{4 \text{ m}} \\ B_y &= 14,43 \text{ kN}.\end{aligned}$$

So

$$A_y = 20,62 \text{ kN} - 14,43 \text{ kN} = 6,19 \text{ kN}.$$

Q1.1	$(\bar{x}; \bar{y}; \bar{z}) = (-2; 0,28; 2) \text{ m}$
Q1.2	See Figure 0.1.
Q1.3	$\mathbf{u}_{T_1} = 0,666\mathbf{i} + 0,333\mathbf{j} - 0,666\mathbf{k}$ $\mathbf{u}_{T_2} = 0,6\mathbf{j} - 0,8\mathbf{k}$
Q1.4	$T_1 = 24,74 \text{ kN} \quad T_2 = 20,62 \text{ kN}$ $A_x = -16,49 \text{ kN} \quad A_y = 6,19 \text{ kN}$ $A_z = 32,99 \text{ kN} \quad B_y = 14,43 \text{ kN}.$

RESULT

Problem 2

Consider a steel T-section cantilever beam loaded as shown in Fig. 2.

Let $L = 3\text{ m}$, $l = 1\text{ m}$, $w_0 = 0,5 \frac{\text{kN}}{\text{m}}$, $F_0 = 3\text{ kN}$, $E = 200\text{ GPa}$, $t = 10\text{ mm}$, $f = 25\text{ mm}$, $b = 100\text{ mm}$, $h = 300\text{ mm}$.

- Q2.1 List all kinematic constraints (boundary conditions) and the associated reactions, determine the degree of static indeterminacy and compute the reaction forces.
- Q2.2 Plot the shear force and bending moment diagrams.
- Q2.3 At the cross section located at $x = L/2$, determine the state of stress at point C and compute the absolute maximum shear stress (hint: draw the three Mohr circles at point M)
- Q2.4 Determine the elastic curve (derive an expression for the beam deflection); in particular, compute the deflection at point C.

The fixed support is now replaced by a pin support, and a column is added below the beam, connected by a pin at point B. The column is a circular cylinder of radius R , length l and is made of the same material as the beam.

- Q2.5 Assuming a safety factor of 1,5, find the minimum radius R , of the column required to prevent buckling while it supports the beam.

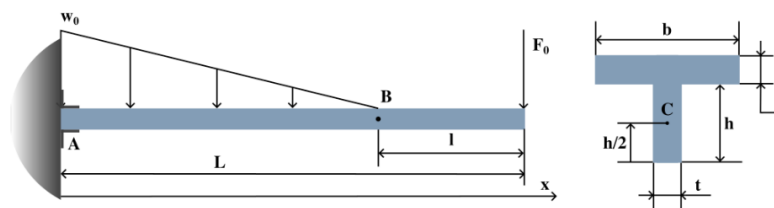


Fig. 2. A steel T-section cantilever beam.

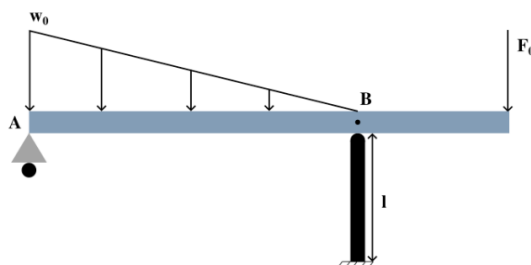


Fig. 3. A simply supported steel T-section beam.

Derivation

- Q2.1 For the structure, which is a cantilever beam fixed at A we have that transverse deflection and rotation are fixed, $v(0) = 0$ and $v'(0) = 0$. At the free end no external moment is applied, so the internal bending moment here must be zero, $M(L) = 0$. As we have an applied shear here, the internal shear at the end must be $V(L) = F_0$. This means we have three unknown reactions, A_x , A_y and M_0 and three equilibrium equations $\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$ which means the structure is determinate.

Q2.2 The distributed load corresponds to a point load of

$$W = w_0 \cdot (L - l) \cdot \frac{1}{2} = 0,5 \frac{\text{kN}}{\text{m}} \cdot 2 \text{ m} \cdot \frac{1}{2} = 0,5 \text{ kN}.$$

Which acts at a $1/3$ of the distance $L - l$, i.e.

$$x_W = \frac{1}{3} \cdot (L - l) = 0,666 \text{ m}.$$

Using a force equilibrium in the y direction we obtain the reaction A_y at the wall as

$$\sum F_y = 0$$

$$A_y - W - F_0 = 0$$

$$A_y = W + F_0$$

$$A_y = 0,5 \text{ kN} + 3 \text{ kN} = 3,5 \text{ kN}.$$

Therefore our shear diagram will start at $V(0) = 3,5 \text{ kN}$. Therefore, the shear diagram is given by:

$$V(x) = \begin{cases} A_y - \int_{L-l}^x w(s) ds & 0 \leq x \leq L-l \\ A_y + W & L-l \leq x \leq L. \end{cases}$$

We know that $w(s) = \frac{w_0}{2l}(L-l-s)$ so using GeoGebra we can plot the entire shear diagram as shown on Figure 0.2

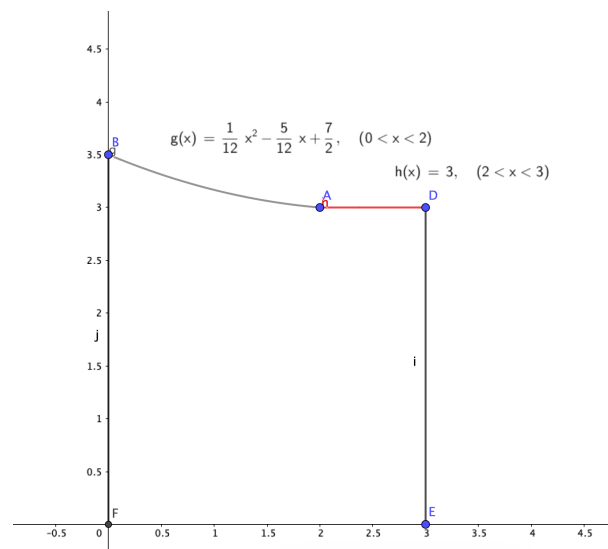


Figure 0.2: Shear diagram

From GeoGebra we got the definition

$$V(x) = \begin{cases} \frac{1}{12}x^2 - \frac{5}{12}x + \frac{7}{2} & 0 \leq x \leq L-l \\ 3 & L-l \leq x \leq L. \end{cases}$$

We now use

$$M'(x) = V(x) \implies M(x) = \int V(x) dx.$$

and integrate piecewise to get:

$$M(x) = \frac{1}{36}x^3 - \frac{5}{24}x^2 + \frac{7}{2}x + M_0, 0 \leq x \leq L-l.$$

Using a moment equilibrium about point A we can find M_0 as:

$$\sum M_A = 0$$

$$M_0 + W \cdot x_W + F_0 \cdot L = 0$$

$$M_0 = -W \cdot x_W - F_0 \cdot L$$

$$M_0 = -0,5 \text{ kN} \cdot 0,666 \text{ m} - 3 \text{ kN} \cdot 3 \text{ m}$$

$$M_0 = -9,33 \text{ kN}.$$

For the last integration we simply get:

$$M(x) = 3x + C_2, L - l \leq x \leq L.$$

For continuity at $x = L - l$ we require

$$\frac{1}{36}8 - \frac{5}{24}4 + \frac{7}{2}2 - 9,33 = 3 \cdot 2 + C_2 \quad = \frac{1}{36} \cdot 8 - \frac{5}{24} \cdot 4 + \frac{7}{2} \cdot 2 - 9,33 - 6 = -8,94111.$$

Hence,

$$M(x) = \begin{cases} \frac{1}{36}x^3 - \frac{5}{24}x^2 + \frac{7}{2}x - 9,33 & 0 \leq x \leq 2 \\ 3x - 8,94 & 2 \leq x \leq 3. \end{cases}$$

This is shown on [Figure 0.3](#).

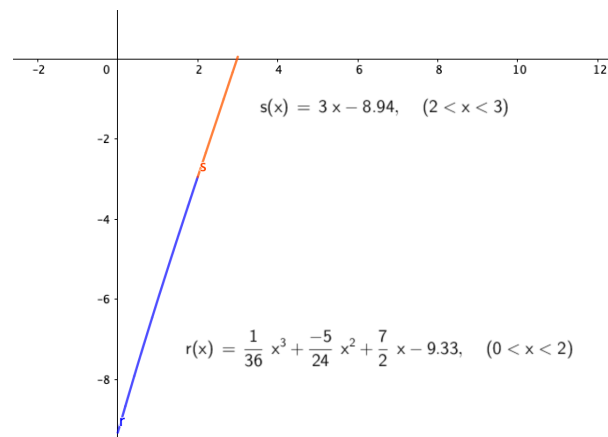


Figure 0.3: Moment diagram

Q2.4 For an Euler-Bernoulli beam, we have that:

$$v''(x) = \frac{M(x)}{EI}.$$

Which will be integrated piecewise. For $0 < x < L - l$ we get

$$\begin{aligned} v_1''(x) &= \left(\frac{1}{36}x^3 - \frac{5}{24}x^2 + \frac{7}{2}x - 9,33 \right) \cdot \frac{1}{EI} \\ v_1'(x) &= \frac{0,02778 \left(\frac{x^4}{4} - 2,5x^3 + 63x^2 - 335,88x \right)}{EI} + C_2 \\ v_1(x) &= \frac{0,001389x^5 - 0,0173625x^4 + 0,58338x^3 - 4,66537x^2}{EI} + C_2x + C_3. \end{aligned}$$

Here we use that $v(0) = 0 \implies C_3 = 0$ and $v'(0) = 0 \implies C_2 = 0$ so

$$v_1(x) = \frac{0,001389x^5 - 0,0173625x^4 + 0,58338x^3 - 4,66537x^2}{EI}.$$

For $L - l < x < L$ we get:

$$\begin{aligned} v_2''(x) &= \frac{3x - 8,94}{EI} \\ v_2'(x) &= \frac{3 \left(\frac{x^2}{2} - 2,98x \right)}{EI} + C_4 \end{aligned}$$

$$v_2(x) = \frac{0,5x^3 - 4,47x^2}{EI} + C_4x + C_5.$$

For continuity at $x = L - l$ we require

$$v_1'(L - l) = v_2'(L - l)$$

$$v_1(L - l) = v_2(L - l).$$

So:

$$\frac{0,02778 \left(\frac{(2)^4}{4} - 2,5(2)^3 + 63(2)^2 - 335,88x \right)}{EI} = \frac{3 \left(\frac{(2)^2}{2} - 2,98 \cdot 2 \right)}{EI} + C_4$$

$$C_4 = \frac{0,02778 \left(\frac{(2)^4}{4} - 2,5(2)^3 + 63(2)^2 - 335,88 \cdot 2 \right)}{EI} - \frac{3 \left(\frac{(2)^2}{2} - 2,98 \cdot 2 \right)}{EI}$$

$$C_4 = -\frac{0,225413}{EI}.$$

and

$$\frac{0,52^3 - 4,472^2}{EI} - \frac{0,225413}{EI} \cdot 2 + C_5 = \frac{0,0013892^5 - 0,01736252^4 + 0,583382^3 - 4,665372^2}{EI}$$

$$C_5 = \frac{0,0013892^5 - 0,01736252^4 + 0,583382^3 - 4,665372^2}{EI} - \frac{0,52^3 - 4,472^2}{EI} + \frac{0,225413}{EI} \cdot 2$$

$$C_5 = \frac{6,08121}{EI}.$$

So:

$$v(x) = \begin{cases} \frac{0,001389x^5 - 0,0173625x^4 + 0,583338x^3 - 4,66537x^2}{EI} & 0 \leq x \leq L - l \\ \frac{0,5x^3 - 4,47x^2}{EI} - \frac{0,225413}{EI}x + \frac{6,08121}{EI} & L - l \leq x \leq L. \end{cases}$$

This is plotted on [Figure 0.4](#).

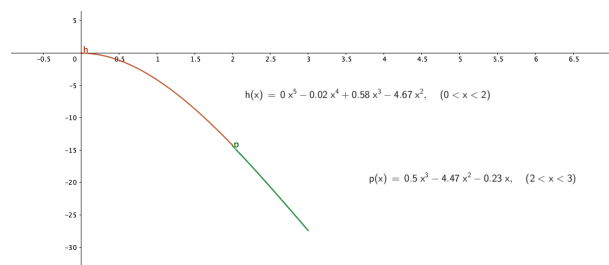


Figure 0.4: Deflection diagram

Using this, the deflection at C is $v(C) = v(L - l) = -14,33 \text{ mm}$.

Q2.5 I do not have time for this part but I would use the formula for critical load:

$$P_{\text{cr}} = \frac{\pi^2 EI}{(KL)^2}.$$

With $K = 0,7$ for a column fixed at the bottom and free at the top. I would then solve for the required second moment of area I and find the radius of the smallest circular rod having at least this high a value of I .

Q2.1	See the section
Q2.2	See Figure 0.2 and Figure 0.3.
Q2.3	No Answer
Q2.4	See Figure 0.4 $v(C) = -14,33 \text{ mm}$
Q2.5	See the section..

RESULT